



CIVITAS indicators

Greenhouse gases tank-to-wheel emissions (ENV_DC_CE1)

DOMAIN

				
Transport	Environment	Energy	Society	Economy

TOPIC

Decarbonisation

IMPACT

Transport greenhouse gases emissions

Reducing the greenhouse gases emissions of urban mobility

ENV_DC

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Transport activity is a major contributor to pollution, significantly impacting air quality, human health, and climate change. A substantial portion of transport activity takes place in urban areas, where high population density and concentrated economic activity lead to high travel demand. The reliance on motorized vehicles powered by fossil fuels exacerbates climate change by emitting greenhouse gases, such as CO₂. By addressing urban transport emissions, cities can enhance air quality and reduce carbon footprints.

This indicator is an estimation of the amount of urban mobility greenhouse gases emissions. **It is a relevant indicator when the policy action is aimed at reducing the impact of urban mobility and transport on climate. A successful action is reflected in a LOWER value of the indicator.**

DESCRIPTION

The indicator is the quantity of tank-to-wheel GHG emissions expressed in **tonnes/year**.

METHOD OF CALCULATION AND INPUTS

There are three methods for the calculation of the indicator. Method 1 relies on the number of vehicle-kilometres travelled in the pilot area collected through survey or travel diary. This method has relatively low to moderate complexity, depending on the availability and quality of the survey or travel diary data. A single survey may collect data to calculate multiple indicators beyond this one. The significance of the results depends strongly on the representativeness and accuracy of the collected data. Well-designed surveys can provide robust estimates of travel activity.

Method 2 consists of extracting the value of emissions from an urban transport model. **If a city transport model exists** and provides an estimation of the greenhouse gas emissions from all relevant modes of transport, this method has zero complexity. Furthermore, assuming that the model is reliable, results have a good level of significance. However, **if an urban model does not exist**, its complexity becomes high because building an urban transport model is not straightforward.

Method 3 estimates greenhouse gas emissions building on energy use for urban transport activity. The energy use should be available by fuel type. Energy use by fuel type can be obtained from different sources. One option is collecting energy supplied at refuelling stations and energy operators (see factsheet related to the indicator ENG_EF_ED1). Another option is estimating energy used building on vehicle fleets, mileage and use factors (see factsheet related to the indicator ENG_EF_ED3). A third option is extracting energy consumption from an urban transport model, if there is a city transport model providing energy consumption by fuel type (but not greenhouse gas emissions, otherwise the first method can be applied). This method is not very complex; more complex than extracting emissions from an existing model, but definitely less complex than building an urban transport model to estimate emissions. Its level of significance depends on the completeness of the data on energy use. On average, method 3 is probably less significant than method 2.

METHOD 1	METHOD 2	METHOD 3
Tank-to-wheel CO₂ emissions based on distances travelled	Tank-to-wheel greenhouse gas emissions drawn from an urban transport model	Tank-to-wheel greenhouse gas emissions estimated from energy consumption by fuel type
Complexity 	Complexity 	Complexity 
Significance 	Significance 	Significance 

Method 1

Calculation of CO₂ emissions based on distances travelled

Significance: **0.50**



INPUTS

The following information is needed to compute the indicator:

- a) **The average number of vehicle-kilometres travelled per day in the pilot area by car and by moped/motorbike.** For cars, vehicle-kilometres must be disaggregated by fuel type: respondents should report, along with the kilometres travelled per day by car, the fuel type of their vehicle, so that distances travelled can be attributed to the fuel categories. The data must cover the following fuel technologies:
- Petrol
 - Diesel
 - LPG
 - CNG
 - Non-plug-in hybrid
 - Plug-in-hybrid
 - Battery electric
 - Fuel cells

The data should be collected from a sample survey or travel diary and **scaled to the number of residents and/or users of the pilot area.**

The experiment would result in a modification of the number of vehicle-kilometres per vehicle type and fuel technology.

METHOD OF CALCULATION

The indicator is computed **by the supporting tool** based on the number of daily vehicle-kilometres travelled by cars and mopeds/motorbikes. The user **inputs the daily vehicle-kilometres per vehicle type and fuel technology in the pilot area** into the supporting tool. The tool then calculates the value of the indicator according to the equation and parameters reported in the section below.

EQUATIONS

The value of the indicator is **computed by the supporting tool** as:

$$CO2day = \sum_v \sum_f VKT_{v,f} * EF_{v,f}$$

Where:

$VKT_{v,f}$ = Average number of daily vehicle-kilometres travelled by vehicles of type v and fuel technology f . These are to be obtained by scaling up the survey results to approximate the number of kilometres travelled by all residents and/or users of the pilot area.

$EF_{v,f}$ = Tank-to-wheel CO₂ emissions per vehicle-kilometre by vehicle of type v and fuel technology f . These parameters are predefined values in the supporting tool and presented in the following table.

Vehicle type	Fuel type	CO ₂ Emission factors [kgCO ₂ /km]
Car	Petrol	196
	Diesel	181
	LPG	176
	CNG	169
	Mild hybrid	196
	Plug-in hybrid	137
	Battery electric	0
	Fuel cells	0
Moped/motorbike	Petrol	88

Source: own elaboration based on the EMEP/EEA Air Pollutant Emission Inventory Guidebook 2023 (<https://www.eea.europa.eu/en/analysis/publications/emep-eea-guidebook-2023>).

Emission factors are calculated using the Tier 1 method (Section 1.A.3.b.i–iv), as the product of fuel consumption per vehicle and fuel type (Table 3-15) and the corresponding fuel-specific CO₂ emission factors (Table 3-12).

Emission factors for plug-in hybrid vehicles are approximated as 70% of those of petrol vehicles, reflecting the average share of electric traction over a trip reported in the source. Mild hybrid vehicles are assumed to have the same emission factors as internal combustion engine vehicles using the same fuel type.

Method 2

Tank-to-wheel greenhouse gas emissions drawn from an urban transport model

Significance: **0.50**



INPUTS

The following information is needed to compute the indicator:

- b) **The volume of tank-to-wheel greenhouse emissions** in the period covered by the urban transport model.

Urban transport models can refer to different periods e.g., one peak hour, two peak hours, one peak and one off-peak hour, an average day and so on. Whatever is the period, the greenhouse emissions provided by the model are the input required, which need to be translated in an annual value (see Method of Calculation). If the model already provides annual emissions, this information is already the indicator.

The experiment would result in a modification of the amount of greenhouse gas emissions estimated by the model.

METHOD OF CALCULATION

The indicator should be computed **exogenously** according to the following steps:

- **Identification of the period covered by the transport model** (e.g., one peak hour, two peak hours, whole day, etc.).
- **Identification of the share of daily of traffic covered by the transport model.** The share of traffic covered by the transport model depends on the modelled period and on the distribution of transport activity in the 24 hours. This distribution is different in different contexts. If a transport model exists, this parameter is usually known. If not, 10% is a reasonable value for a morning peak hour.
- **Definition of the factor to extrapolate from day to year.** Most of the models refer to an average working day. If so, this term depends on the number of working days per year. Again, if a model exists, this parameter is usually known. If not, 270 working days/year can be considered.
- **Calculation of the extrapolation factor.** The extrapolation factor should be the product of two terms above (see the following equations).
- **Application of the extrapolation factor and estimation of the indicator** (see the following equations).

EQUATIONS

The extrapolation factor should be computed as:

$$ExtpFact = \frac{1}{DayModShr} * DaytoYear$$

Where:

DayModShr = share of daily of traffic covered by the transport model

DaytoYear = Factor to extrapolate from day to year

Example, if the model covers two hours and *DayModShr* = 16% of daily traffic, and 270 working days (*DaytoYear*) are considered, the extrapolation factor is:

$$ExtpFact = \frac{1}{0.16} * 270 = 1687$$

The value of the indicator is then computed as:

$$GHGEmis = ModGHGEm * ExtpFact$$

Where:

$ModGHGEm$ = GHG emissions related to the modelled period extracted from the model

Method 3

Tank-to-wheel greenhouse gas emissions estimated from energy consumption by fuel type

Significance: **0.25**



The following information is needed to compute the indicator:

- a) **The amount of energy used in a given period** for each relevant fuel type. The fuel types to be considered are the followings:

- **Gasoline**
- **Diesel**
- **LPG**
- **CNG**

If some of these fuel types are not relevant in the experiment area, they are not considered.

- b) **The carbon content** of each fossil fuel type. This element consists of a set of values based on chemical characteristics of the fossil fuels. The following values can be used unless more specific values based on the method of production can be available in the pilot site:

Gasoline: 2.31 kg/l

Diesel: 2.68 kg/l

LPG: 1.51 kg/kg

CNG: 2.74 kg/kg

The experiment would result in a modification of the amount of energy used. The carbon content of each fuel is fixed.

METHOD OF CALCULATION

The indicator should be computed **exogenously** according to the following steps:

- **Identification of the existing refuelling stations.**
- **Definition of a sample of refuelling stations** if their total number is too large for collecting data from all of them.
- **Collection of data from the refuelling stations**
- **Estimation of total amount of fuel supplied** in case the data is collected from a sample of stations (see equation below).
- **Estimation of the GHG emissions** (see equation below)

EQUATIONS

If the amount of fuel is collected from a sample of refuelling stations, the total value of energy supplied for these two fuel types should be estimated according to the following equation:

$$EngSupl^e = \frac{\sum_s EngSupl_s^e}{n} * N$$

Where:

$EngSupl_s^e$ = Amount of fuel type e supplied by sampled refuelling station s in the monitored four weeks

n = Number of sampled refuelling stations

N = Total number of refuelling stations supplying fuel type e in the urban area

The GHG emissions can be obtained by simply applying carbon content factors to the total value of energy by fuel type:

$$GHGEmiss = \sum_e (EngSupl^e * CarbCont^e)$$

Where:

$CarbCont^e$ = the carbon content factor of fuel type e

ALTERNATIVE INDICATORS

This indicator refers to tank-to-wheel greenhouse emissions from transport in the pilot environment. Non-fossil fuels are not considered in this indicator. Since greenhouse gas emissions have a broad impact, rather than a local one, it is reasonable arguing that the production of non-fossil fuels (especially electricity) should be considered. In that case, indicator **ENV_DC_CE2** should be considered. That indicator is however more complex, because it requires either a more sophisticated existing urban model dealing with well-to-tank emissions or more extended data on energy use by fuel type as well as on the mix of fuels used to produce electricity.